

Am I My Brother's Barkeeper? Sibling Spillovers in Alcohol Consumption at the Minimum Legal Drinking Age*

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Abstract

We use data on sibling pairs near the minimum legal drinking age to provide causal estimates of peer effects in alcohol consumption. Following prior work on other outcomes, we exploit the discontinuous increase in alcohol consumption of the older sibling at the legal drinking age in a regression discontinuity design. Our preferred point estimates imply that the number of binge drinking days reported by the younger sibling *decreases* by 27% of the mean at the cutoff. While our estimates are somewhat imprecise, we are consistently able to rule out positive estimates from the existing literature. Our research design provides estimates which are interpretable as the causal effect of the peer's alcohol consumption. This is in contrast to most prior work which instead identifies the causal effect of exposure to the peer. We explain how this distinction matters for policy.

Keywords: Alcohol, Peer effects, Siblings, Minimum legal drinking age

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1 Introduction

Excessive alcohol consumption has been shown to harm young adults in a variety of dimensions including health, educational performance, criminal activity, and criminal victimization (Carpenter and Dobkin, 2009; Carrell et al., 2011; Lindo et al., 2013; Carpenter and Dobkin, 2015, 2017; Chalfin et al., forthcoming). Among the many potential determinants of adolescent alcohol use, peer effects have received substantial attention from academics and policymakers.¹

Leading empirical work provides strong evidence of positive peer effects in alcohol consumption via random assignment to college roommates (Duncan et al., 2005; Eisenberg et al., 2014; Guo et al., 2015). However, this design exploits variation in exposure to *a peer*. These results do not tell us what peer behavior or characteristic explains the spillovers—they are “contextual” as opposed to “endogenous” peer effects, where the latter is an effect driven specifically by peer alcohol consumption (Manski, 2000). Distinguishing between these two types of peer effects has key implications for policy. With endogenous peer effects, an intervention that influences the alcohol consumption of recipients also has an effect on the consumption of their untreated peers. This is not necessarily the case under contextual effects. Further, this design cannot be replicated in many other peer groups of interest, notably siblings—a peer group that has been the focus of a substantial and rapidly growing literature on a wide range of outcomes.²

This paper provides causally interpretable estimates of peer effects in alcohol consumption between siblings. We focus on siblings residing in the same household and exploit a discontinuous increase in older sibling alcohol consumption at the minimum legal drinking age (MLDA) using a regression discontinuity design (RDD).³ Our results are, to our knowledge, the first quasi-experimental estimates of peer effects in alcohol consumption between siblings. Further, our estimates are interpretable as the causal effect of sibling alcohol consumption. This extends the prior literature on peer effects in this behavior more generally, which primarily estimates the causal effect of exposure to heavier drinking roommates (who may have other unobserved characteristics that influence peer alcohol consumption).

¹In its Underage Drinking Fact Sheet, the National Institute of Alcohol Abuse and Alcoholism lists peer pressure as one of three key causes of alcohol consumption among adolescents (<https://pubs.niaaa.nih.gov/publications/UnderageDrinking/UnderageFact.htm>).

²Including, for example, risky behaviors (Duncan et al., 2001; Fagan and Najman, 2005; Harris and López-Valcárcel, 2008; Altonji et al., 2016), health (Breining, 2014; Ho, 2017; Cawley et al., 2019; Daysal et al., 2019), fertility (Heissel, 2021), education (Goodman et al., 2015; Joensen and Nielsen, 2018; Karbownik and Özek, 2019; Altmejd et al., 2021), and military service (Bingley et al., 2021).

³An identification strategy first used by Carpenter and Dobkin (2009) to study the effect of alcohol consumption on mortality.

Attempts to estimate causal peer effects face three main difficulties. First, the “reflection” problem implies that a simple regression of peer A’s behavior on peer B’s cannot determine the direction of any observed effect (Manski, 1993). Second, peer groups are (typically) endogenous, which is a concern if individuals choose peers who have similar preferences related to the behavior of interest. Finally, individuals who share a peer group are likely to experience unobservable common shocks which are correlated with their outcomes.

Our setting and identification strategy address each of these difficulties. By focusing on siblings, a peer group which is naturally occurring, we avoid the potential for selection into the peer group. By restricting the sample to siblings with different ages, we ensure that the variation in alcohol consumption that we utilize is both exogenous (avoiding the problem of common shocks) and specific to one of the two siblings (avoiding the reflection problem).

To estimate spillover effects in alcohol consumption we utilize data from the 1997 National Longitudinal Survey of Youth (NLSY97), which has three key features for our analyses. First, the data contains unique “household roster” information on all individuals living with the NLSY97 respondent. These rosters include the birth months of each household member and their relationship to the respondent, which we use to construct our running variable. Second, it contains several dimensions of alcohol consumption for survey respondents that are measured at relatively high frequency (consumption in the past month). Third, all NLSY97 respondents were between the ages of 12 and 17 during the first wave of the survey, which ensures that many survey respondents will have older siblings near the age 21 cutoff.

Our main results estimate the effect of the oldest sibling’s legal access to alcohol on the younger (second oldest) sibling’s alcohol consumption.⁴ In the full sample of 4,278 sibling-pair-years, our estimates imply a *negative* relationship between the older sibling’s legal access to alcohol and the younger sibling’s alcohol consumption but are typically somewhat imprecise. In our preferred specification we estimate that the number of binge drinking days (5 or more drinks) in the past 30 days reported by younger siblings decreases by 0.34 days at the cutoff. This is a substantial effect given that the average younger sibling in the sample reports 1.14 binge drinking days in the past month. While this estimate is statistically significant (95% CI of [-0.650, -0.025]), similar specifications which differ in terms of outcomes or control variables are often less precise. However,

⁴In most cases we only observe alcohol consumption for one sibling in a household (the sibling that is a NLSY97 respondent). However, we do estimate the discontinuity in older sibling alcohol consumption in a separate sample of NLSY97 respondents who have younger (usually not NLSY-respondent) siblings. Results are similar to the existing literature utilizing this research design, typically implying increases in consumption of 20-30%.

we do consistently find a moderate negative effect across a wide range of specifications. Further, our preferred estimate is well into the left tail of a distribution of placebo discontinuities estimated at older sibling ages that are close, but not equal to, 21 years.

These estimates are almost always precise enough to rule out the relatively large positive effects reported in prior work. Notably, we compare our estimates to those from Eisenberg et al. (2014) (henceforth EGW) since they utilize what is arguably the gold-standard research design (randomized college roommates) on this question, and their results are smaller than nearly all other estimates in the related literature.⁵ EGW’s key comparable result is that being assigned a roommate who binge drank in 30 days before move-in leads to a 19% increase in the probability of any binge drinking by the respondent in a second 30-day period roughly 8 months later. After scaling our preferred estimate for the same outcome by the control mean, we obtain a 95% CI of [-27.6%, +9.2%]. Further, among 64 total specifications estimated for this outcome, 59 confidence intervals (95%) exclude an effect of +19%. It is worth emphasizing again that we *do not* view these results as contradictory since EGW estimate contextual and not endogenous peer effects.

As a whole, we view our results as suggestive evidence that the causal relationship in alcohol consumption between siblings in this age range is negative. This is admittedly counterintuitive. However, a large literature has established that a series of negative outcomes spike at the MLDA (e.g., Carpenter and Dobkin, 2017, 2015; Lindo et al., 2013; Carrell et al., 2011; Carpenter and Dobkin, 2009) and it is plausible that younger siblings update their beliefs about the costs of alcohol consumption after observing their older siblings experience these negative consequences.⁶ Further, the wider literature on peer effects has often considered the potential for spillovers to operate through similar pathways related to “expectations” about a behavior.⁷ Unfortunately, our data does not allow us to directly test for evidence of this mechanism,⁸ but we do find that our effects vary in several key dimensions in ways that are consistent with a non-zero effect.

First, our estimates are especially negative for measures of excessive alcohol consumption (binge drinking) on the intensive margin. Prior work suggests that negative effects of alcohol consumption

⁵Notably Gaviria and Raphael (2001), Duncan et al. (2005), Lundborg (2006), Fletcher (2012), Guo et al. (2015), and Altonji et al. (2016) all report similar or larger peer effects of alcohol consumption in their preferred specifications.

⁶Some specific negative outcomes that have been studied in the literature are likely too rare to explain our spillover effects. However, despite the lack of quantitative evidence it is likely that other less-severe and more common negative outcomes also spike at age 21.

⁷See, e.g., the categorization of peer effect mechanisms in Manski (2000).

⁸Both the NLSY97 data and several other related datasets, notably the National Health Interview Survey (NHIS) and the National Longitudinal Study of Adolescent Health (Add Health) do contain related info. However, the NLSY97 only asks these questions in a small number of survey waves (restricting our sample size), the NHIS has few respondents under age 21, and Add Health lacks the detailed household information necessary to construct our running variable.

experienced at the MLDA are driven by these types of consumption (Carpenter et al., 2016), and it is reasonable to expect that any spillover effects would be most apparent on this margin as well. Second, our estimates are more negative (and statistically significant) among siblings who are the same gender. It is reasonable to assume that these sibling pairs would be more exposed to one another, and that any true spillover effect would be stronger in this group. Third, our estimates are more negative when the older sibling is a male—this fits with prior literature which finds that males report larger increases in alcohol consumption, and especially excessive alcohol consumption, at the MLDA (Carpenter et al., 2016).

Three specific concerns which may threaten the interpretation of our results are worth highlighting. First, an older sibling’s legal access may directly affect the access of the younger sibling (e.g., if the older sibling purchases alcohol for the younger). However, this effect would bias our negative estimates toward zero.⁹ Second, parents may compensate for an expected spillover effect with a change in parenting style (e.g., by adopting a stricter parenting style) when the older sibling turns 21, which would likely negatively bias our results. The NLSY97 includes information on parenting styles which allows us to provide evidence against this concern.¹⁰ Third, it is possible that younger siblings in our sample are not sufficiently aware of the alcohol consumption of their older siblings (e.g., if they do not spend much time together). To further support our focus on this peer group we use the American Time Use Survey (ATUS)—which includes a minute-level breakdown of what activities were performed during a single day, where, and with whom—to demonstrate that siblings in the relevant age range who live together spend substantial amounts of time together.

We contribute to three separate literatures. First, we add to a limited number of studies which provide quasi-experimental estimates of endogenous peer effects in alcohol consumption. The most similar work is Fletcher and Marksteiner (2017), who estimate endogenous peer effects in alcohol consumption with a different identification strategy (randomized assignment to an alcohol cessation program) in a very different population (adult spouses in which one spouse is a heavy drinker). Second, we provide what we believe to be the first quasi-experimental estimates of spillovers in alcohol consumption between siblings. The existing economics literature on this topic has focused on

⁹Further, if an access effect exists it would likely be driven by an increase in older sibling consumption. In this scenario the access effect would be part of our parameter of interest (the endogenous peer effect) and would not bias our results away from the true value of that parameter.

¹⁰Even in the presence of these confounders our results are interpretable as the spillover effects of the older sibling’s legal access to alcohol which is itself policy relevant given contemporary debates on lowering the MLDA (Carpenter and Dobkin, 2011; DeJong and Blanchette, 2014). See, e.g. <https://www.usatoday.com/story/news/nation-now/2017/11/09/drinking-age-19-these-wisconsin-lawmakers-aim-change-law/849365001/> for details on a MLDA change recently under debate in the Wisconsin state legislature.

other outcomes (e.g., smoking as in Harris and López-Valcárcel (2008)) or has relied on structural assumptions (Altonji et al., 2016), while a substantial literature outside of economics has documented the strong correlation in consumption between siblings (e.g., Duncan et al., 2001; Fagan and Najman, 2005; Trim et al., 2006; Van Der Vorst et al., 2007; Whiteman et al., 2013). Third, we contribute to a growing literature on sibling spillovers more generally, where a growing body of evidence has demonstrated the importance of sibling influences in a variety of important domains including health outcomes (Breining, 2014; Ho, 2017; Cawley et al., 2019; Daysal et al., 2019), fertility (Heissel, 2021), education (Goodman et al., 2015; Joensen and Nielsen, 2018; Karbownik and Özek, 2019; Altmejd et al., 2021), and military service (Bingley et al., 2021).

The rest of this article is structured as follows. [Section 2](#) describes the data. [Section 3](#) details the methods used in the analysis. [Section 4](#) describes the results, and [Section 5](#) concludes.

2 Data

2.1 NLSY97

The NLSY97 is a longitudinal survey of 8,984 American youths who were between the ages of 12 and 17 at the time of the first wave of the survey in 1997 (Bureau of Labor Statistics, 2019).¹¹ Designed and administered by the Bureau of Labor Statistics, the NLSY97 includes a wide variety of questions related to family processes, education, employment, health, and family formation, among other topics. The NLSY97 has three key features which make it uniquely suitable for our analysis.

First, information on the alcohol consumption of respondents is obtained in all waves. Specifically, we use the following outcomes, all reported for the past 30 days prior to the survey: the number of days on which any drinks were consumed (drinking days), the number of days on which 5+ drinks were consumed (binge days), and the binary versions (any/none) of drinking days and binge days.¹² Second, the survey includes detailed information on the composition of each respondent’s household, including the relationship of each household member to the respondent, and the age of each household member (including the respondent) in months.¹³ Third, each wave includes

¹¹Waves are annual from 1997-2011, and biennial beginning in 2013.

¹²Additional measures of alcohol consumption used to define subgroups, but not used as outcomes, include whether or not the respondent had ever consumed alcohol as of the previous year and whether or not the respondent had ever binge consumed alcohol as of the previous year.

¹³Specifically, the public use version of the NLSY97 includes the exact date of the interview along with the month of birth for the respondent and all members of the respondent’s household. This allows us to estimate the age in months of the respondent and all siblings who ever co-resided with the respondent during the survey, regardless of

a detailed set of covariates at the respondent and household levels which are also potentially related to alcohol consumption and therefore will serve as control variables in our analysis. These include the race and gender of the respondent and siblings, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year.¹⁴

For reasons that we will describe in the next section, we will primarily focus on 2,614 NLSY respondents who have only one older sibling in their household, and where that older sibling is between the ages of 19 years & 0 months (228 months of age) and 23 years & 0 months old (276 months of age). Alternative samples drop either the requirement that the siblings *currently* live together,¹⁵ the requirement that the two siblings are the oldest two siblings, or both. Key subgroups of interest are defined based on parental educational attainment, whether the siblings report the same gender, the gender of the older sibling, the age difference between the siblings, and the younger sibling’s previous (as of the prior survey wave) drinking experience. Summary statistics are shown in [Table 1](#).

The NLSY97 sampling process first identified a large number of households and then identified all individuals in each household who were age-eligible for the survey. Because of this, a sample of siblings who are all NLSY97 respondents is available. While this sample will not be used in our main analysis, we do use it to provide descriptive evidence of between sibling correlation in alcohol consumption in the NLSY97. [Table 2](#) shows results from a regression of the younger sibling’s past month alcohol consumption on the older sibling’s past month alcohol consumption for multiple measures of consumption in a sample of sibling pairs that are both NLSY97 respondents.¹⁶ These results are not causally interpretable and are not directly comparable to our sample of interest. However, they show that in this sample, much like in prior literature described previously, there is a very strong correlation in alcohol consumption between siblings.

Among a set of measures of past month alcohol consumption in the NLSY97, we focus on whether those older siblings are themselves NLSY97 respondents. The NLSY97 also collects a very limited set of information on close relatives such as siblings who live outside of the respondent’s household. This information is unfortunately too limited to make use of these siblings in our analyses.

¹⁴This is not an exhaustive list of potentially relevant control variables included in the NLSY97. A large number of relevant questions in the NLSY97 are either asked too intermittently or have too many missing values to be useful in our analysis.

¹⁵They must have lived together during at least one NLSY97 wave in order for us to observe the sibling and the sibling’s age in months.

¹⁶Households with 3 or more sibling NLSY97 respondents are excluded for simplicity and the sample is limited to individuals under the age of 23.

measures of the excessive alcohol consumption in our main analyses: The number of binge drinking days (5+ drinks) in the past month, and an indicator for any binge drinking days in the past month. This helps make our results more comparable to the existing literature, which typically focuses on measures of binge drinking, and more policy relevant, since excessive drinking is more likely to result in the various negative outcomes that are associated with alcohol consumption. The focus on a small subset of the available outcomes also helps to reduce problems of multiple testing. However, results for all outcomes are presented for certain analyses in order to demonstrate robustness.

2.2 ATUS

In a supplementary analysis, we use the ATUS to provide descriptive evidence that siblings in the relevant age range constitute an important peer group (Hofferth et al., 2020). The ATUS is a repeated cross-sectional survey, which regularly samples a subset of Current Population Survey (CPS) respondents and solicits a detailed time-diary for one day’s worth of activities. These time diaries detail the minute-level activities of respondents, including what activity was being performed, where, and with whom. Since the ATUS sample is drawn from the CPS, the two surveys can be linked so that the household structure of each ATUS respondent is observable. We use this information to select all respondents from the 2003-2019 waves of the ATUS who lived with at least one older sibling who was 23 years old or younger. We then use the information on who activities were performed with, to characterize the strength of these sibling relationships and provide suggestive evidence regarding the potential for younger siblings to be exposed to changes in older sibling alcohol consumption. Summary statistics for our analysis sample of 3,158 ATUS respondents are shown in [Table 3](#).

3 Methods

We implement a reduced form regression discontinuity design in which the age of the older sibling is the running variable and the effect of the older sibling turning 21 (gaining legal access to alcohol) acts as an instrument for the older sibling’s alcohol consumption. The MLDA was first used as an exogenous source of variation in alcohol consumption in a RDD by Carpenter and Dobkin (2009) to study the effect of alcohol consumption on mortality and has since been used to study a

wide range of other outcomes.¹⁷ So long as no other factors related to the outcome are changing discontinuously at the cutoff, this approach will provide causally interpretable results. Many of the aforementioned studies have provided convincing evidence of both a strong first stage (large, discontinuous increases in consumption at age 21) and of the credibility of this research design (e.g., establishing that observable covariates do not change discontinuously at the cutoff).

Many NLSY97 respondents have more than one older sibling. As pointed out by Dahl et al. (2014), this complicates estimation of peer effects with a RDD since it is unclear how to define the running variable. To avoid these complications, we define the peer group as siblings who currently reside in the same household and consider the effect of the oldest sibling in the peer group on the second oldest sibling. This ensures that we have a large sample of sibling pairs in which we observe the information necessary to implement the RDD (older sibling age in months and younger sibling alcohol consumption) and allows us to avoid the complications involved with defining the running variable in peer groups with more than two members.

The reduced form RD is then implemented by estimating the following equation via OLS:

$$alc_{2,h,t} = \gamma_1 1\{age_{1,h,t} \geq 21\} + f(age_{1,h,t}) + X'_{2,h,t}\gamma_2 + X'_{1,h,t}\gamma_3 + W'_{h,t}\gamma_4 + \theta_{2,h} + \mu_{2,h,t} \quad (1)$$

Where subscripts denote sibling (2 = younger, 1 = older), household (h), and the survey wave (t), the outcome is some measure of the younger sibling's past month alcohol consumption, the reduced form effect of older sibling's legal access to alcohol on the younger sibling's consumption is estimated by γ_1 , $f(age_{1,h,t})$ is a flexible polynomial in the running variable (e.g., older sibling age fully interacted with the cutoff dummy $1\{age_{1,h,t} \geq 21\}$), $X_{2,h,t}$ is a vector of younger sibling covariates, $X_{1,h,t}$ is a vector of older sibling covariates, $W_{h,t}$ is a vector of household h level covariates, and $\theta_{2,h}$ is a younger sibling fixed effect.

We separately estimate the increase in older sibling drinking at age 21 in a similar equation (where the outcome is the alcohol consumption of the older sibling, $alc_{1,h,t}$, and younger sibling characteristics are removed from the regression). The sibling pairs used in that analysis are not the same as those used to estimate the reduced form effect.¹⁸

¹⁷Including criminal activity (Carpenter and Dobkin, 2015; Hansen and Waddell, 2016), crime victimization (Chalfin et al., forthcoming), morbidity (Carpenter and Dobkin, 2017), marijuana consumption (Yörük and Yörük, 2011; Crost and Guerrero, 2012; Crost and Rees, 2013; Yörük and Yörük, 2013), the consumption of other illegal drugs (Deza, 2015), risky sexual behavior Yörük and Yörük (2015), and mental health (Yörük and Yörük, 2012).

¹⁸Sibling pairs in the reduced form sample are those in which the second oldest sibling in the household is a NLSY97 respondent. Sibling pairs in the first stage are those in which the oldest sibling is a NLSY97 respondent. Although a two-sample two-stage least squares is theoretically possible, sample sizes are likely to be too small for it to be informative.

To construct the running variable, we use the birth month of the older sibling and the month of the relevant interview (younger sibling’s interview in the reduced form, older in the first stage). This implies that our running variable is rounded *up*, and that the cutoff indicator will be mismeasured for some sibling pairs in which the older sibling is exactly 21 years (252 months) old. Following the recommendations in Dong (2015), we address this misclassification bias with a “donut RD” specification that excludes sibling pairs in which the older sibling is exactly 21 years (252 months) old. Following the existing literature on MLDA-based RDDs, the sample is limited to sibling pairs in which the older sibling is between the ages of 19 and 23—i.e., the youngest older sibling in our sample is 19 years and 0 months old (228 months of age) and the oldest is 23 years and 0 months old (276 months of age). Standard errors are cluster robust at the younger sibling level to account for correlation in the error term among observations from same individual.

Additional models test the sensitivity of our results in several of the dimensions mentioned above (bandwidth, correction for rounding-induced bias, inclusion of covariates, inclusion of fixed effects, and order of the running variable polynomial). Notably, we implement the continuity-based RD framework developed by Cattaneo et al. (2019), including mean-squared error optimal bandwidth selection and bias-corrected robust confidence intervals.

The main assumption required for a causal interpretation of γ_1 is that no unobserved confounding factors change discontinuously at the cutoff. We “test” this assumption to the extent possible by estimating models similar to equation 1, where outcomes are predicted values from separate regressions which predict the alcohol consumption measures using a range of covariates. A second required assumption in RDDs more generally is that the running variable is not manipulated. Although this is not technically a concern in our setting (since age is not manipulable), a related problem can arise if rates of nonresponse to alcohol consumption questions change discontinuously at the cutoff. This is primarily a concern in the first stage where an older sibling may be more willing to report alcohol consumption once they reach age 21. We test this assumption by demonstrating visually that the density of older sibling age is smooth through the cutoff and by testing formally for a discontinuity in the density at the cutoff.

4 Results

4.1 Design

We begin by demonstrating that older sibling drinking changes discontinuously at age 21. Results using count and binary versions of drinking and binge days in the past month are shown in [Table 4](#). Similar to results from previous work using the same identification strategy in different samples,¹⁹ there is a large discontinuous increase in alcohol consumption at age 21. This effect is apparent in all models shown in [Table 4](#). In preferred models (donut specification with a linear function of the running variable, controls,²⁰ and individual level fixed effects), past month binge drinking and drinking days increase by 0.45 days and 1.45 days at the cutoff. Both increases are statistically significant at the 5% level. Increases on the extensive margin of consumption for binge drinking and drinking (from the same models) are 5.7pp and at 8.3pp respectively, which are both statistically significant at the 5% level. [Figure 1](#) plots mean values of our four outcomes among older siblings in each month-of-age bin along with fitted lines and regression coefficients from the corresponding first stage regressions. These graphs reinforce the results summarized from [Table 4](#).

[Figure 2](#) graphs the distribution of the older sibling’s age in months, for the first stage sample. While age is not manipulable, it is possible for nonresponse to change discontinuously at the cutoff and such a result could potentially affect the interpretation of the first stage results presented in [Table 4](#). [Figure 2](#) suggests that this is not occurring, given that the distribution is relatively smooth through the cutoff at 21 years of age.

In [Table 5](#) and [Figure 3](#) we present results which test for discontinuities in sibling pair characteristics at the cutoff that could potentially confound the causal effect of interest. We use a set of covariates²¹ to predict each of our four alcohol consumption outcomes and then use a regression analogous to our reduced form specification to test for a discontinuity in each of those predictions at the cutoff (standard errors are calculated via bootstrap). All discontinuities are statistically indistinguishable from zero.

¹⁹See the previous section for citations.

²⁰Month and year of the survey, the race and gender of the respondent and siblings, the age of the respondent, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year.

²¹Specifically: age, gender of both siblings, educational enrollment, highest completed education, work status, indicators for whether the household lives in an urban area, census region dummies, AFQT score, household size, and interview month/year. (All variables refer to the younger sibling’s information unless otherwise noted.)

4.2 Reduced form

Table 6 shows results from equation 1, estimated in the full sample of sibling pairs who reside in the same household and are the oldest siblings in the household, for all four measures of alcohol consumption. Focusing on the preferred model (the first column) which includes the previously defined vector of controls and individual level fixed effects, all point estimates are negative and large (e.g., younger sibling past month binge days decrease by 0.337 day or 26.9% of the mean just before the cutoff). Assuming that no unobserved confounder changes discontinuously when the older sibling turns 21, this implies that the older sibling’s increase in alcohol consumption at the cutoff has a causal negative effect on the alcohol consumption of younger siblings on this margin.

Point estimates are statistically significant for only one of the four considered outcomes in our preferred specification, suggesting that these estimates should be interpreted with caution. However, the outcome for which our results are strongest (count of binge drinking) is also arguably the most important—as it represents a particularly risky and problematic type of consumption. Further, point estimates for the preferred models are negative and at least modestly sized for all outcomes in the preferred specification, and nearly all estimates in Table 6. As a percentage of the mean of the outcome just before the cutoff, coefficients in the preferred specification imply that binge drinking decreases by 9.2% at the extensive margin, while drinking (any amount) decreases by 5.9% at the intensive margin and 6.7% at the extensive margin.

Corresponding binned scatter plots for all four outcomes are shown in Figure 4. These figures include coefficients from a regression of each outcome variable on the age of the older sibling in months (centered at 21 years old), an indicator for older sibling age ≥ 21 , and their interaction. The figures show that there is no significant change in the alcohol consumption of younger sibling when their older sibling turns to 21, suggesting (as also shown in Table 6) that the size and precision of our negative estimates are somewhat sensitive to specification choices.

Why might an increase in older sibling alcohol consumption at the MLDA have negative effects on younger sibling consumption? One natural interpretation is that younger siblings learn about the costs of alcohol consumption by observing the experiences of their older siblings. While the NLSY97 data does not allow us to test for this mechanism directly (e.g., by using some measure of knowledge or beliefs about the effects of alcohol consumption as an outcome in our RDD), we argue that a true non-zero effect is likely to be stronger in certain subgroups than others, discussing this point and relevant subgroup estimates in subsection 4.3. Further, heterogeneity across different

outcomes (described above) is also consistent with this proposed mechanism. Our results imply that younger siblings reduce consumption at all margins that we test. However, these responses are strongest for the margins (binge drinking, especially at the intensive margin) that are most risky and that have been found in the prior literature to drive negative outcomes at the MLDA (Carpenter et al., 2016).

To further strengthen our results, we perform a series of placebo tests estimating discontinuities at older sibling age values near but not equal to 21. Specifically, for each monthly age bin up to 12 months on either side of the cutoff, we re-estimate equation 1. Results are plotted in Figure 5, with the estimate at the actual cutoff value represented by a blue vertical line. While this test cannot be used to conduct formal inference, the results are reassuring. The true estimate is in the far left tail of both placebo distributions.

4.3 Heterogeneity across subgroups

To determine which subgroups may be driving the relationships seen in the full sample, we re-estimate our reduced form specification in several subgroups of interest. Specifically: parental education (all parents completed high school vs. one or more did not complete high school), whether the two siblings report the same or different genders, the gender of the older sibling, the age difference between the siblings ($\geq$ vs. $<$ the median of 30 months), and the younger sibling’s experience of drinking or binge drinking as of the prior year’s survey (any vs. none).

We show our main results from this analysis in Table 7. Our specifications align with the preferred model from Table 6 (a linear donut RD specification with controls and younger sibling FEs included) and the outcome is the number of binge drinking days in the past month. In Table 8 we report the same results from similar regressions with an indicator for any binge drinking as the outcome. Several of these subgroups are defined based on covariates which could reasonably be expected to mediate the sibling spillover effect of interest—either because the groups identify siblings who are likely more exposed to each other (gender match, age), proxy for heterogeneity in the first stage (older sibling gender (Carpenter et al., 2016)), or identify younger siblings with different levels of prior (pre-cutoff) knowledge about alcohol.

Point estimates are typically negative and large but not statistically significantly different from zero. One key exception is the subgroup where both siblings report the same gender. In that subgroup, younger sibling consumption decreases by a statistically significant 0.616 binge drinking days when the older sibling turns 21. Intuitively, a more pronounced effect in this subgroup makes

sense if we expect that siblings of the same gender are likely to spend more time together. We also find that effects are strong when the older sibling is a male. In this subgroup, younger siblings report 0.662 less binge drinking days in the past month at the cutoff. An estimate which is again statistically significantly different from zero. This result is reassuring given estimates in the prior literature which suggest that the MLDA has a larger effect on the drinking behavior of males. We estimate similarly heterogeneous effects in these subgroups with the binary outcome in [Table 8](#), although all estimates there are less precise.

Patterns for subgroups defined by prior drinking experience, and age difference, are less clear. One possible explanation for this is that these characteristics may be strongly correlated with each other, or with other key factors which predict alcohol consumption. If the age difference between siblings is large, then the younger sibling is likely to have less prior alcohol consumption simply because they are younger. This complicates the interpretation of these subgroup estimates. In [Table 7](#), point estimates are noticeably more negative among siblings with larger age differences. Does this reflect the age difference? Or the fact that younger siblings in such pairs have less drinking experience? Further, less prior drinking experience for one sibling likely implies the same for the other (due to the strong positive correlation in consumption between siblings), creating a similar complication. If we observe smaller discontinuities among younger siblings with less prior drinking experience, is that evidence against an expectations pathway (since such younger siblings should have “more to learn”)? Or does that estimate simply reflect a smaller discontinuity in older sibling alcohol consumption?

4.4 Comparison to prior literature

While we believe that the negative effects we estimate are plausible, our sample sizes are relatively small and the estimates are not always statistically significantly different from zero. However, given the uniformly positive effects estimated in the related literature (e.g., Gaviria and Raphael, 2001; Duncan et al., 2005; Clark and Lohéac, 2007; Lundborg, 2006; Fletcher, 2012; Eisenberg et al., 2014; Guo et al., 2015; Altonji et al., 2016; Fletcher and Marksteiner, 2017) a null result is interesting in its own right.

We simplify our comparison to the prior literature by focusing on Eisenberg et al. (2014) (EGW). We choose this study for two reasons. First, their estimates are smaller than nearly all other estimates in the related literature. Therefore, if our null results are sufficiently precise to rule out

the EGW point estimate, the same is true for a series of other leading estimates in the literature.²² Second, EGW utilize what we see as the best existing identification strategy on this question: randomized assignment to college roommates.

EGW’s key comparable result is that being assigned a roommate who binge drank in 30 days prior to a baseline survey fielded in August of 2009 (just before the students moved in to their shared rooms) leads to a 19% increase in the probability of any binge drinking by the respondent in a second 30-day period roughly 8 months later. In [Figure 6](#) we scale our preferred estimate for the same outcome by the control (immediately pre-cutoff) mean, obtaining a 95% CI of [-27.6%,9.2%] in our preferred specification and ruling out the EGW point estimate (red line) in all four specifications. In the right-hand panel, we repeat this comparison for a series of our subgroup estimates in our preferred specification. Here 95% CIs exclude the EGW point estimate in most subgroups.

Since we estimate the direct effect of the peer’s alcohol consumption, while EGW estimate the effect of exposure to a peer who binge drank in the past, we do not view these results as contradictory. Instead, if college roommates and co-resident older siblings at similar ages are similar peer influences, these estimates suggest that the effects estimated by EGW result at least in part from roommate characteristics other than drinking behavior. As mentioned previously, this is critically important for policy. The presence of endogenous peer effects that result directly from the peer’s alcohol consumption indicate that the costs of alcohol consumption, and the benefits of consumption-reducing interventions, are multiplicative.

Although the EGW treatment (a roommate who did not binge drank vs. one who did) may seem larger than the effects of the MLDA on older-sibling alcohol consumption, timing implies that the difference is small. The EGW treatment is a 100% increase in exposure peer binge drinking measured roughly 8 months in the past, while we measure a contemporaneous spillover effect. Especially given the trajectory of alcohol consumption in this age range (as demonstrated by various figures in this paper), it is likely that many of the “control” peers in EGW (no prior binge drinking) would have binge drank by the time the outcomes were measured 8 months later. Therefore, the contemporaneous treatment vs. control difference in peer binge drinking in the EGW sample is likely much smaller than 100%.

²²Notably Gaviria and Raphael (2001), Duncan et al. (2005), Lundborg (2006), Fletcher (2012), Guo et al. (2015), and Altonji et al. (2016) all report similar or larger peer effects of alcohol consumption in their preferred specifications.

4.5 Evidence against offsetting parental responses

Parental responses are one key pathway through which the older sibling’s legal access to alcohol could directly (i.e., not via an endogenous peer effect) affect the younger sibling’s consumption of alcohol. The NLSY97 includes two separate questions which can provide some suggestive evidence on the role that parental responses may or may not play in explaining the results described above. First all respondents are asked to score the degree to which they are monitored by their parents (including, if applicable, parents they live with and parents they do not live with) on a scale from 0-16, with higher scores indicating closer monitoring. Second, respondents are asked to classify their parents’ parenting styles (again, including if applicable parents who do and do not live with the respondent) as either uninvolved, permissive, authoritarian, or authoritative.

In the first panel of [Table 13](#) we present results for models similar to equation 1 with the parental monitoring score for the parents that the respondent lives with. If the respondent lives with two parents, the outcome is their average score. In the second panel of [Table 13](#), similar results are presented with an indicator for whether or not at least one parent was reported to be either authoritarian or authoritative.

The survey questions underlying these outcomes are asked only in survey rounds between 1997 and 2000. Luckily, a substantial portion of the sibling-pair-years that meet the sample restriction requirements are in this time period. However, some are not, and sample sizes are therefore smaller than the previously described full sample results in [Table 6](#). Models in [Table 13](#) are the same as those in [Table 6](#) in order to demonstrate the robustness of the results.

In each of the 5 models for each of the three outcomes, point estimates for a discontinuity in parental behavior at the cutoff are either negative (suggesting parents become less strict or engage in less monitoring), not statistically significant, or both. In the preferred model (1) neither point estimate is economically significant. We interpret this as suggestive evidence that parental responses are not driving the observed decrease in younger sibling alcohol consumption at the cutoff.

4.6 Time use of siblings near the MLDA

A concern in our setting is that younger siblings may not be sufficiently exposed to changes in older sibling alcohol consumption that occur at the MLDA. This might be the case if, for example, siblings in this age range do not spend meaningful amounts of time together. Our emphasis on “null” results magnifies the importance of this concern for two reasons. First, prior literature

focuses on different peer groups. If those peer groups are more relevant for alcohol related behavior of adolescents, it is not surprising that we would find smaller effects in our setting. Second, if the siblings in our sample do not spend meaningful amounts of time together, there would be no opportunity for younger siblings to observe, and be affected by, the behavior of their older sibling.

While a small literature on the shared time use of siblings does exist (e.g., Dunifon et al., 2017; Wikle and Hoagland, 2020), there does not appear to be any work on the age ranges relevant for our analysis. We therefore address this concern with a simple descriptive analysis which demonstrates that siblings at ages near the MLDA spend substantial amounts of time together.

Using the ATUS sample described previously (3,158 ATUS respondents who live with an older sibling age 23 or younger), we calculate average time spent per day with an older sibling. For comparison, we also calculate average time spent per day with friends.²³ Finally, we present these means in several different subsets of activities. First, we move from all activities reported in the time-diary to activities performed while not at work or school—under the assumption that such “discretionary” time is more relevant for the behavior we are interested in. Second, we further restrict our focus to time spent not at work or school and without any parents present for the same reason. Finally, we break down this last category of discretionary and unsupervised time into the main activity groups defined by ATUS. Results are shown in [Figure 8](#).

The average respondent reports spending roughly 1 hour and 50 minutes with an older sibling during their diary-day, just under one hour of which is discretionary and unsupervised time, when alcohol consumption would be most likely to occur. Since a substantial portion of the average diary-day in our sample is spent either at work, at school, or with parents (see [Table 3](#)), this is a meaningful amount of time. Moreover, the activities that siblings engage in together during this discretionary and unsupervised time (right panel of the figure) are broadly similar to the activities engaged in with friends. While it is clear that respondents spend more time with friends (especially within discretionary and supervised time), we feel that this is strong evidence that the sibling pairs we study in our main analysis are a meaningful peer group for the behavior of interest.

4.7 Robustness

Our main analyses focus on sibling pairs who currently reside in the same household, are the oldest two siblings in that household, and where the older sibling is between the ages of 19 and 23. In

²³We consider an activity to be performed with an older sibling (or similarly, with a friend) if the so-called “who” record in the ATUS data lists at least one older sibling (or friend).

this subsection, we explore the robustness of our results to different sample selection criteria, and to various specification choices such as the inclusion of controls or FEs.

4.7.1 Alternative samples

Table 9 presents results that remove either the requirement that the siblings currently live together, the requirement that the siblings are the oldest two siblings in the household, or both. The first panel corresponds to **Table 6**, which uses our main sample. Point estimates are broadly consistent (small, negative, not always statistically significant) across these samples but are closer to zero in the alternative samples. We view this as reassuring since a true spillover effect is likely to be larger when the siblings live together (siblings who do not live together likely spend less time with each other) and when the peer is the only older sibling present (if there are multiple older siblings the behavior of any one older sibling may become less influential).

4.7.2 Comparing many specifications

We visually summarize the overall robustness of our results in a plot of 64 specifications (most of which have been presented in other parts of this paper), which vary by the inclusion of fixed effects, the inclusion of controls, the order of the running variable polynomial, and the sample, in **Figure 9** and **Figure 10**. Each regression in **Figure 9** uses an indicator for any binge drinking as the outcome, and each coefficient is scaled by the corresponding “control” mean (mean of the outcome just below the cutoff)—facilitating comparisons with the EGW point estimate of +19%. Among 64 total specifications, 58 scaled estimates are negative, though the magnitudes vary and confidence intervals usually do not rule out a null effect. Among 64 total specifications, 59 confidence intervals (95%) exclude the EGW point estimate. **Figure 10** shows similar results with the count of binge drinking days as the outcome.

One additional reassuring pattern in this figure is that the handful of outlier CIs (in both figures) which include large positive effects *all* include quadratic polynomials in the running variable. Visual inspection of various binscatters in this paper suggest that the relationship between the outcome and running variable is likely linear in our data. Further, higher-order polynomials in RD designs are known to increase the risk of detecting spurious effects (Gelman and Imbens, 2019).

4.7.3 Fixed effects

Figure 9 and (to a lesser extent) Figure 10 highlight that our results are somewhat sensitive to the inclusion of younger sibling FEs. Models with FEs generally have more negative point estimates and tighter confidence intervals. While the improved precision is expected, sensitivity of the point estimates to the inclusion of FEs may appear surprising given that FEs should not be necessary for identification in our research design, and that adding FEs does not change the estimation sample (as shown in, e.g., Table 6). However, as explained in detail by Miller et al. (forthcoming), FE estimates are identified only by groups that have variation in treatment (here, by younger siblings who show up in our sample at least once before their older sibling turns 21 and at least once after).

In Table 10 we demonstrate that such “switchers” make up roughly two-thirds of the corresponding regression sample used in our main analyses (siblings who live together and are the two oldest siblings in the household), and that these switchers differ meaningfully in some dimensions from the remainder of the sample (“non-switchers”). Given prior results (Table 7 and Table 8) showing that treatment effects in this context are heterogeneous, we take this as suggestive evidence that the sensitivity of our main results to the inclusion of FEs is driven by treatment effect heterogeneity. Since FEs also buy us improvements in precision, and we have no *a priori* reason to prefer estimates for the combined sample over the switcher sample, we consider the FE estimates to be preferred.

Table 11 shows characteristics of similar switcher and non-switcher subsets of the alternative sample that *does not* require the siblings to live together at the time of the sample.²⁴ A common reason for sibling pairs in our main sample to appear only when the older sibling is under 21 (i.e., to be non-switchers) is that the older sibling moves out too soon. It is therefore not surprising that switchers make up a much larger proportion of this alternative sample (without the “same household” restriction). Reassuringly, Figure 9 and Figure 10 also demonstrate that results in this alternative sample are less sensitive to the inclusion of FEs.

4.7.4 Effects on older siblings

Table 12 presents estimates of the effect of *younger* sibling’s access to alcohol on the *older* sibling’s alcohol consumption. Since it is less common for sibling pairs in this age range to live together (here the younger sibling is between the ages of 19 and 23), sample sizes under our typical sample definitions are relatively small in this analysis. For this reason we also present these results with

²⁴The sample used in the bottom panel of Table 9 and denoted as “Sample 4” in Figure 9 and Figure 10.

alternative samples as in Table 9. Compared to Table 9, the sign of these point estimates is less consistent (sometimes negative, sometimes positive) and the estimates are broadly less precise—patterns which are more consistent with a true zero effect. One possible explanation is that the proposed expectations mechanism for the negative effects estimated in our main analysis is less plausible in this context—since older siblings are likely to already have substantial experience with alcohol consumption. We view this as additional suggestive evidence in support of our main results.

4.7.5 Bandwidth and estimation

All results discussed thus far have relied on a bandwidth of 24 months (i.e., restricting the sample to sibling pairs in which the older sibling is between the ages of 19 years and 0 months and 23 years and 0 months). This follows prior literature utilizing MLDA RDDs, which nearly universally chooses this bandwidth (e.g., Carpenter and Dobkin, 2009; Deza, 2015; Hansen and Waddell, 2016; Carpenter et al., 2016). We demonstrate the robustness of our main results to the choice of bandwidth in two ways. First, Figure 7 graphs point estimates and confidence intervals from the preferred model²⁵ for 36 separate bandwidths (single month increments from 12 to 48 months). Results are stable across bandwidth choices, although (expectedly) less precise as bandwidth shrinks. Second, we verify that our results persist when using the “continuity” framework for estimation and inference in RDDs developed by Cattaneo et al. (2020). This approach involves local polynomial estimation of the discontinuity, a data-driven bandwidth selection approach which is mean squared error optimal, and bias-corrected confidence intervals. Table 14 presents the results of the continuity-based RD models with the triangular kernel. Column (1) and (3) use MSE-optimal bandwidths, while column (2) and column (4) use our ad-hoc 24-month bandwidth. All columns include the MSE-optimal estimates and bias-corrected robust 95% confidence intervals where the standard errors are cluster robust at individual level. Consistent to Table 6, all coefficients are negative with moderate magnitudes. For models with individual fixed effects, the coefficients are large and statistically significant compared to models without fixed effects, a pattern also present in Table 6.

5 Conclusions

We focus on a population which allows for the estimation of causally interpretable endogenous peer effects in alcohol consumption under relatively weak assumptions: sibling pairs close to the

²⁵Donut RD specification including a vector of controls, a linear polynomial in the running variable fully interacted with the cutoff dummy, and individual level fixed effects

MLDA in the United States. This setting is helpful for two reasons. First, the peer group (sibling pairs) is not chosen. Second, the average alcohol consumption of young adults has been shown to increase discontinuously at the MLDA. This allows for the use of a RDD. While somewhat imprecise, estimates suggest that alcohol consumption of younger siblings decreases when the older sibling gains legal access to alcohol. We are consistently able to rule out large positive effects reported in the prior literature.

Several limitations are worth mentioning. First, as in any RDD, external validity is a concern. Our results apply only to a certain peer group, adolescent siblings residing in the same household where the older sibling is near the MLDA. However, our results are less limited in this way than a typical MLDA-based RDD, given that the running variable and the outcome are taken from different individuals (ages range from 12-20 years for younger siblings in our sample whose older sibling is within one month of the cutoff). Second, an older sibling’s legal drinking status could *potentially* affect younger sibling consumption indirectly, e.g., via parental responses or access effects. However, one key set of confounders (parental responses) are, to some extent, ruled out based on observable information in the data, and another (access effects) are likely to bias our results towards zero.

Our results emphasize the important distinction between contextual (interpretable as the effect of the peer’s behavior) and endogenous (interpretable as the effect of exposure to the peer) peer effects. A key rationale for policy makers to understand peer effects in alcohol consumption and other costly behaviors, is the potential for the costs of those behaviors—or the benefits of interventions which successfully decrease the prevalence of the behaviors—to be socially multiplied. This will occur only if the peer effect in question is endogenous. Many studies on peer effects, notably the leading work on spillovers in risky health behaviors based on randomly assigned college roommates, are only able to identify contextual peer effects. The existence of contextual peer effects in a population does not necessarily imply the existence of endogenous peer effects and policy makers should exercise caution in acting on results which identify only the former and not the latter.

An important goal for future work will be to understand the differences between the growing set of well-identified results on peer effects in alcohol consumption. The research designs used in our work, Fletcher and Marksteiner (2017), and the series of papers utilizing randomly assigned college roommates (Duncan et al., 2005; Eisenberg et al., 2014; Guo et al., 2015) all differ in at least two key dimensions: The peer group studied and the nature of the variation in alcohol consumption exploited by the researcher. Further, the interpretation of the results differs since studies based on randomly assigned roommates identify only contextual peer effects. Future work should aim

to understand to what extent these three factors explain the differences in these results and to understand the nature of spillovers in alcohol consumption in more generalizable contexts.

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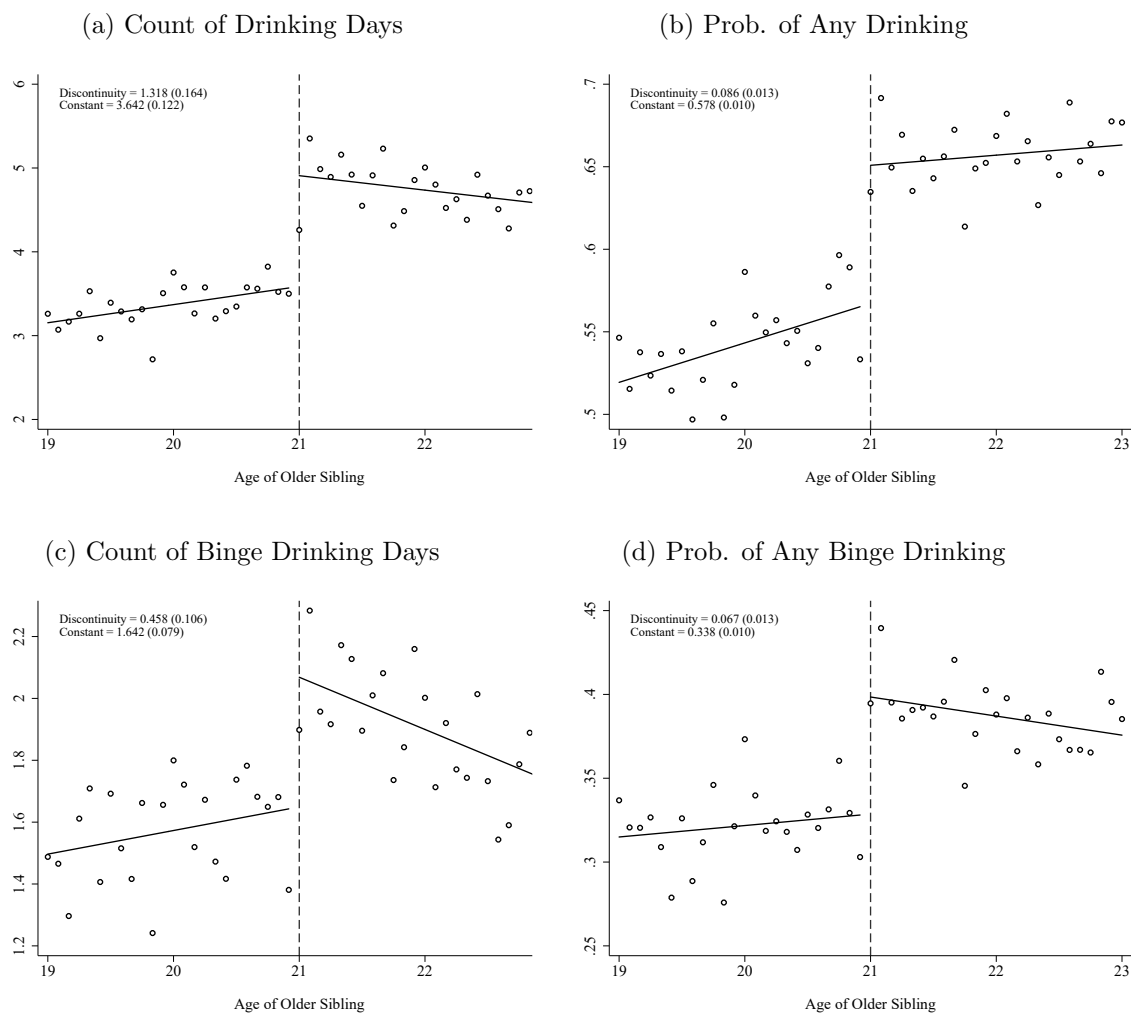
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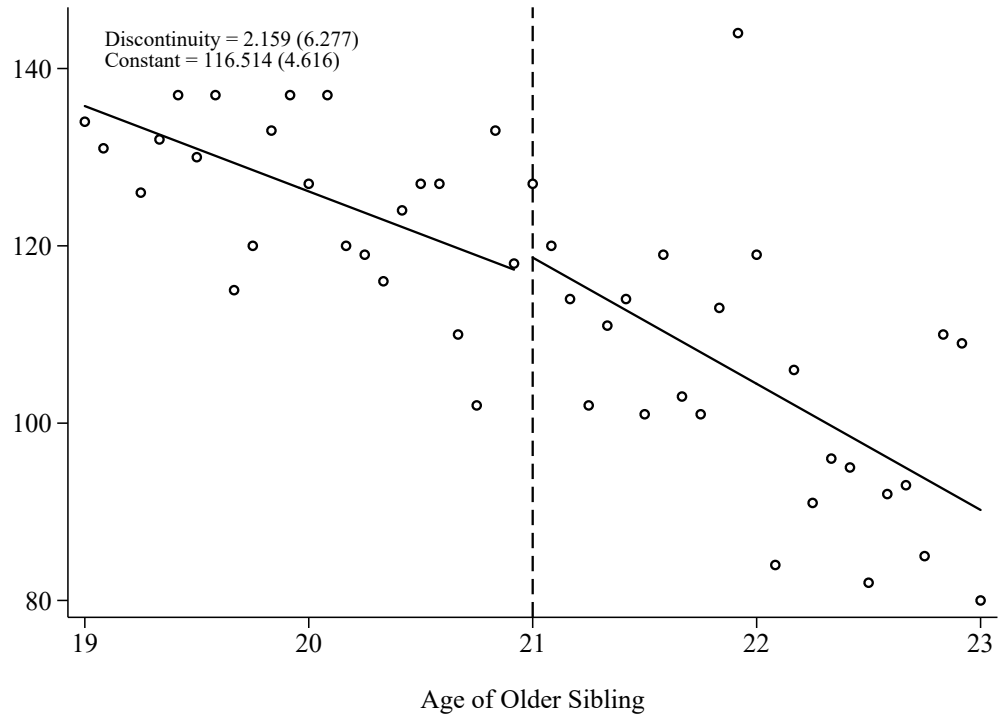
Figures and Tables

Figure 1: Discontinuities in Older Sibling Alcohol Consumption



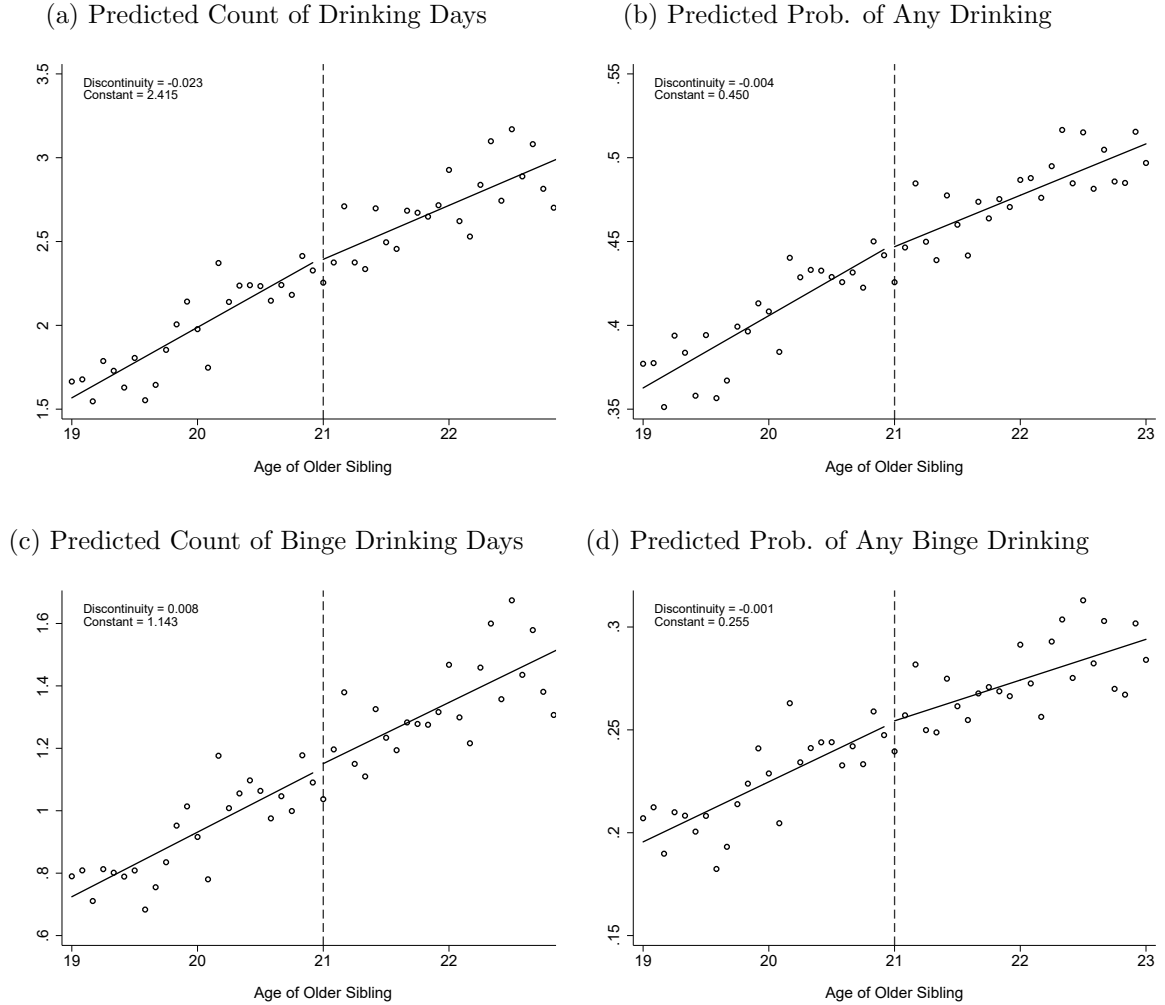
Notes: Each panel shows mean alcohol consumption in each older-sibling-age bin (in months), with linear fits estimated separately on each side of the cutoff. Point estimates and standard errors at the top left of each panel are from the corresponding OLS regressions of the relevant alcohol consumption measure on age (centered at 21), an age-21+ indicator, and their interaction (estimated with individual level data). The sample includes all NLSY97 respondents in the age range who have 1 or more younger siblings and are the oldest siblings in their household.

Figure 2: Density of Running Variable



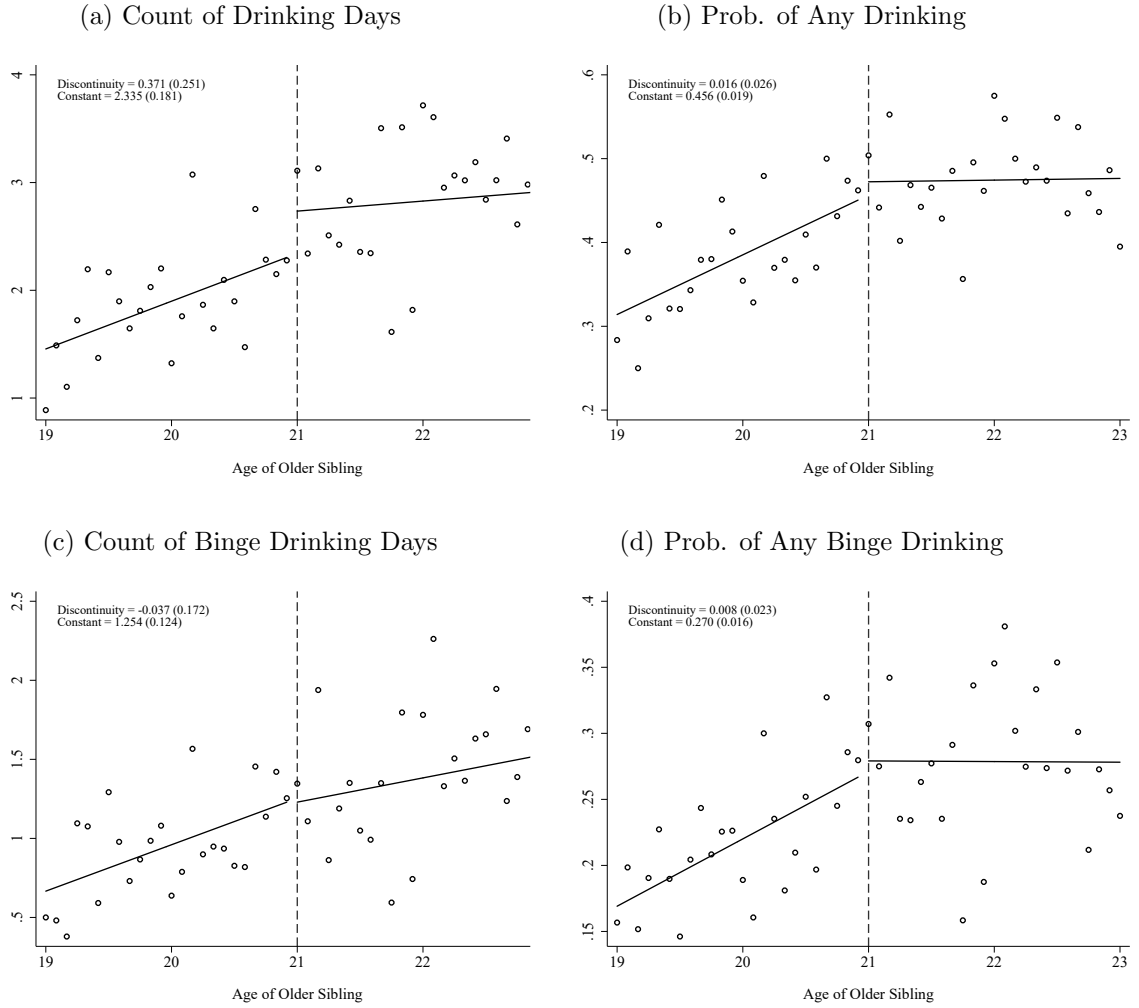
Notes: Y-axis shows the count of sibling-pairs with each running variable value (age in months of older sibling). Point estimates and standard errors are from a regression of the number of observations in each bin on the centered-at-21 running variable, the treatment cutoff, and their interaction (estimated with individual level data).

Figure 3: Smoothness of Covariates at Cutoff



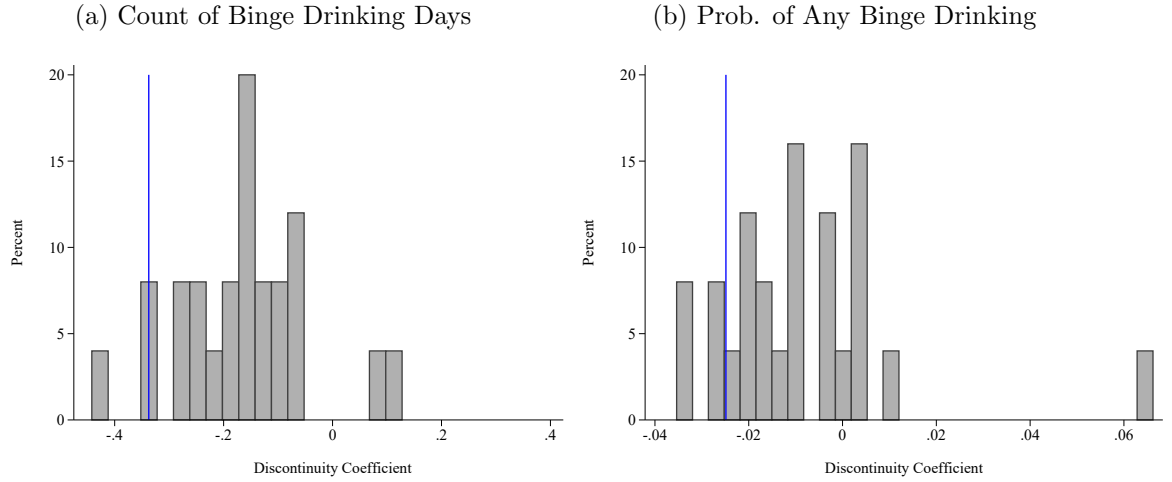
Notes: Each panel shows mean predicted alcohol consumption in each older-sibling-age bin (in months), with linear fits estimated separately on each side of the cutoff (using the binned means). Predicted alcohol consumption measures are fitted values from a regression of the corresponding alcohol consumption measure on age, gender of both siblings, educational enrollment, highest completed education, work status, indicators for whether the household lives in an urban area, census region dummies, AFQT score, household size, and interview month/year. (All variables refer to the younger sibling's information unless otherwise noted.) Point estimates and standard errors at the top left of each panel are from corresponding OLS regressions of the prediction on the running variable, the cutoff indicator, and their interaction (estimated with individual level data). Standard errors are calculated via bootstrap. The sample includes all NLSY97 respondents in the age range who are the second oldest siblings in their household.

Figure 4: Reduced Form



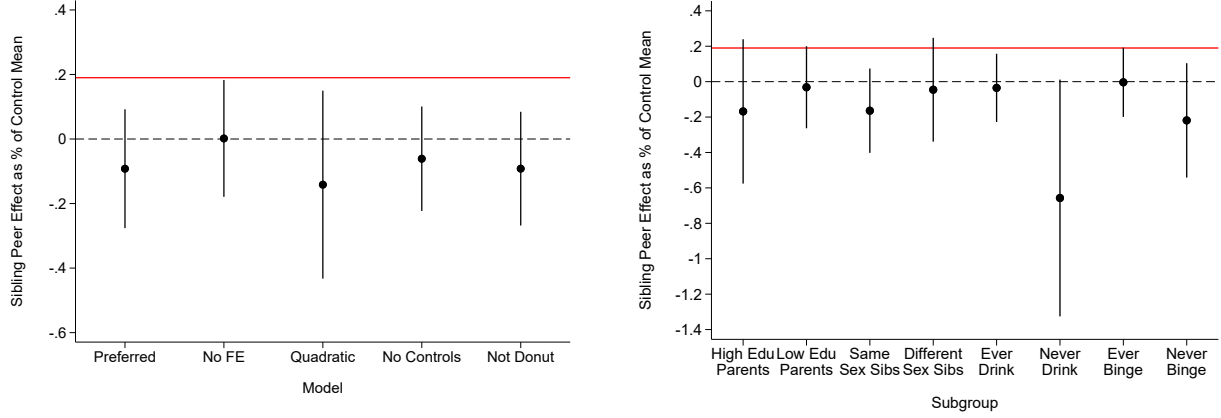
Notes: Each panel shows mean alcohol consumption in each older-sibling-age bin (in months), with linear fits estimated separately on each side of the cutoff. Point estimates and standard errors at the top left of each panel are from corresponding OLS regressions of the younger sibling's alcohol consumption on the running variable, the cutoff indicator, and their interaction (estimated with individual level data). The sample includes all NLSY97 respondents in the age range who are the second oldest siblings in their household.

Figure 5: Placebo Discontinuities



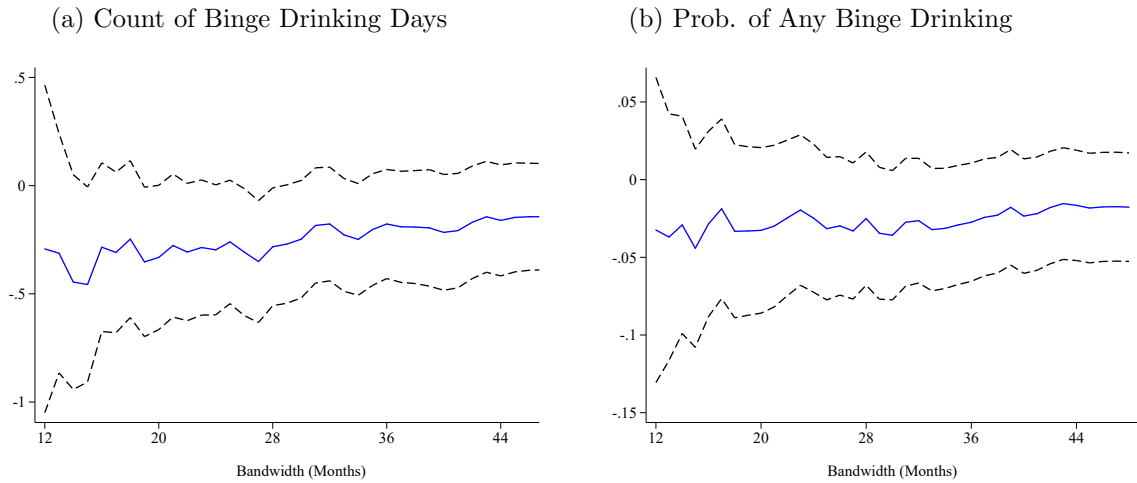
Notes: Figure shows the distribution of point estimates from a series of placebo discontinuity estimates, at running variable values near but not equal to the cutoff. Each underlying regression uses the preferred specification in [Table 6](#) (donut, linear, controls, FE) and a different cutoff age (in months) which varies from one year below to one year above age 21. Actual estimate is shown with a vertical blue line.

Figure 6: Comparison with Eisenberg et al. (2014)



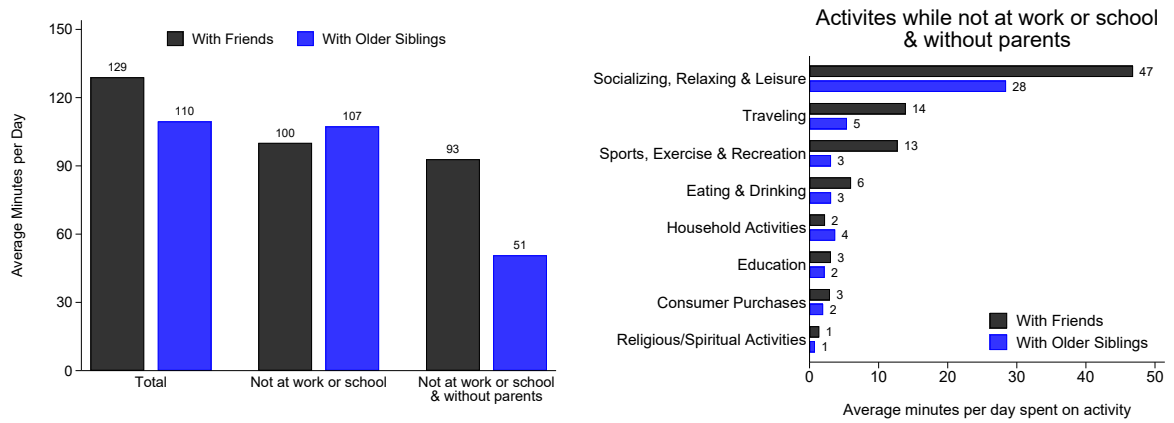
Notes: Figure shows point estimates and confidence intervals from various specifications in Table 6 and Table 8 alongside the main estimate from EGW (red line). The outcome in each regression (including EGW's) is an indicator for any binge drinking days in the past month.

Figure 7: Robustness to Different Bandwidths



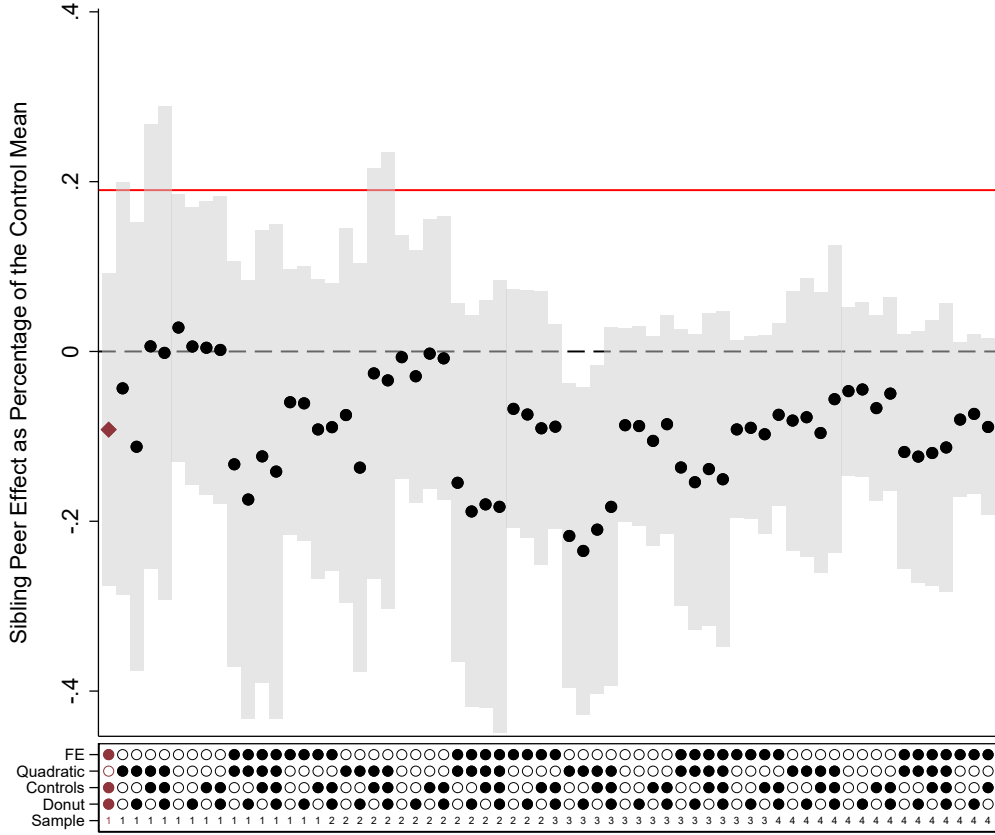
Notes: Figure shows point estimates and confidence intervals from the preferred specification in Table 6 (donut, linear, controls, FE) estimated with different bandwidths (in months).

Figure 8: Time Spent with Older Siblings in the ATUS



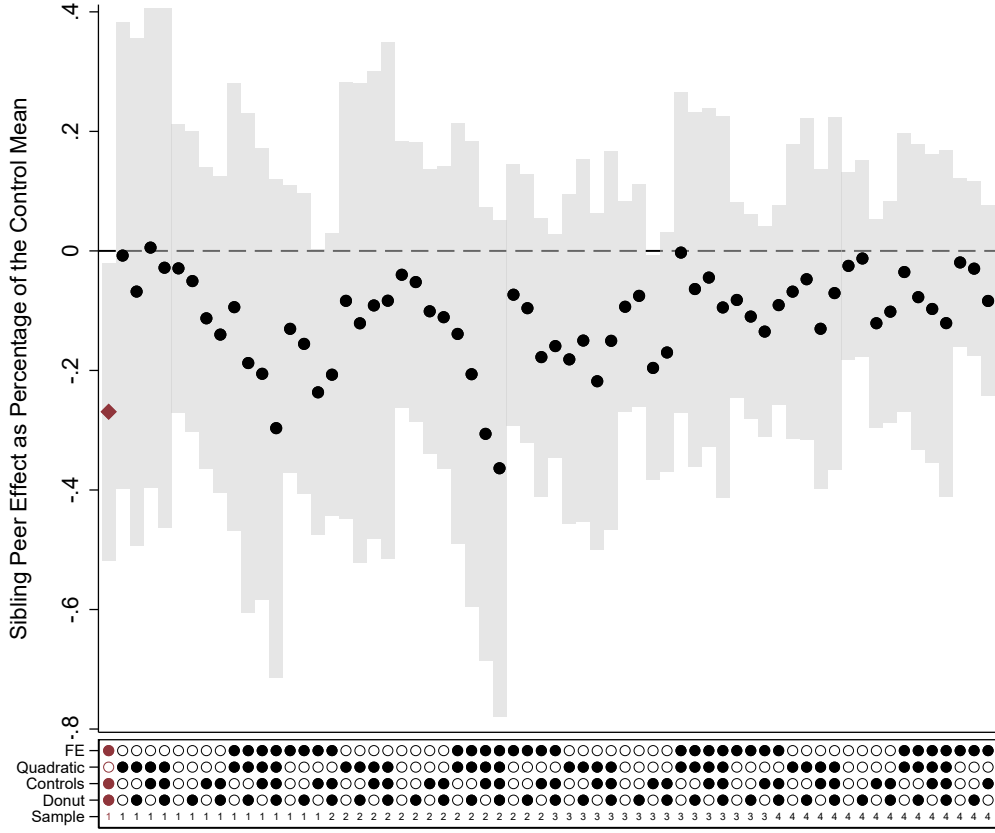
Notes: Descriptive statistics are from a sample of 3,158 respondents in the 2003-2019 waves of the ATUS who have 1 or more older siblings under the age of 23 in their household. Each panel shows the average (across respondent) time per day spent with either friends (black, left bars in each pair in the left panel and top bars in each pair in the right panel) or older siblings (blue, right bars in each pair in the left panel and bottom bars in each pair in the right panel) in various categories.

Figure 9: Robustness to Alternative Specifications
(Outcome = Prob. of Any Binge Drinking)



Notes: Figure shows point estimates and confidence intervals from various specifications in [Table 6](#), all with an indicator for any binge days as the outcome, and all scaled by the pre-cutoff mean (constant in the corresponding regression on the centered-at-zero running variable, the treatment cutoff, and their interaction). Red line shows the point estimate from Eisenberg et al. (2014) for comparison. “Sample” refers to different subsets of sibling pairs, defined by (i) whether they live together, and (ii) whether they are the oldest two siblings in the household (or family). 1 = same household, oldest two siblings in family. 2 = same household, peer is closest older sibling but not necessarily the only older sibling. 3 = siblings do not necessarily live together, siblings are the oldest two siblings in the family. 4 = siblings do not necessarily live together, peer is closest older sibling but not necessarily the only older sibling.

Figure 10: Robustness to Alternative Specifications
(Outcome = Count of Binge Drinking Days)



Notes: Figure shows point estimates and confidence intervals from various specifications in [Table 6](#), all with the count of binge drinking days as the outcome, and all scaled by the pre-cutoff mean (constant in the corresponding regression on the centered-at-zero running variable, the treatment cutoff, and their interaction). “Sample” refers to different subsets of sibling pairs, defined by (i) whether they live together, and (ii) whether they are the oldest two siblings in the household (or family). 1 = same household, oldest two siblings in family. 2 = same household, peer is closest older sibling but not necessarily the only older sibling. 3 = siblings do not necessarily live together, siblings are the oldest two siblings in the family. 4 = siblings do not necessarily live together, peer is closest older sibling but not necessarily the only older sibling.

Table 1: NLSY97 Sample Summary Statistics

	Full Sample mean/sd	RDD Sample mean/sd
<u>Younger Sibling's Past Month Drinking:</u>		
Drinking Days	2.41 (4.98)	2.31 (4.76)
Any Drinking Days	0.41 (0.49)	0.42 (0.49)
Binge Days	1.02 (3.03)	1.14 (3.25)
Any Binge Days	0.23 (0.42)	0.24 (0.43)
<u>Younger Sibling's Characteristics:</u>		
Age	18.72 (3.97)	17.99 (1.72)
Female	0.47 (0.50)	0.48 (0.50)
Race: Black	0.28 (0.45)	0.26 (0.44)
Race: Hispanic	0.26 (0.44)	0.24 (0.43)
In High School	0.53 (0.50)	0.55 (0.50)
In College	0.15 (0.35)	0.20 (0.40)
Urban	0.78 (0.41)	0.77 (0.42)
Worked Past Yr	0.66 (0.47)	0.71 (0.46)
Household Income	\$63,484.53 (58,841.17)	\$61,559.45 (55,247.96)
AFQT score	42,842.18 (28,720.84)	44,616.99 (29,150.67)
<u>Older Sibling's Characteristics:</u>		
Age	22.15 (5.29)	20.87 (1.17)
Female	0.47 (0.50)	0.47 (0.50)
Worked Past Yr	0.38 (0.49)	0.39 (0.49)
Currently Enrolled In School	0.53 (0.50)	0.50 (0.50)
N	15,230	5,702

Notes: The full sample consists of all NLSY respondents with older siblings living in the same household. Our analysis sample ("RDD sample") consists of co-resident sibling pairs who are the oldest two siblings in the household, where the younger sibling is a NLSY97 respondent and the older sibling is between the ages of 19 and 23.

Table 2: Correlations in Alcohol Consumption Between Siblings

	Younger Sibling Consumption			
	Count of Drinking Days		Any Drinking Days	
	(1)	(2)	(1)	(2)
Older Sibling Consumption	0.161*** (0.020)	0.136*** (0.019)	0.178*** (0.018)	0.168*** (0.016)
Constant	3.419*** (0.918)	5.074*** (0.997)	0.485*** (0.113)	0.487*** (0.091)
Mean	1.938	2.356	0.374	0.421
N	3808	6096	3808	6096
	Count of Binge Drinking Days		Any Binge Drinking Days	
	(1)	(2)	(1)	(2)
	(1)	(2)	(1)	(2)
Older Sibling Consumption	0.163*** (0.026)	0.133*** (0.021)	0.166*** (0.018)	0.140*** (0.016)
Constant	2.043** (0.642)	2.874*** (0.604)	0.326*** (0.091)	0.434*** (0.081)
Mean	0.881	1.084	0.198	0.236
N	3806	6096	3806	6096
Lag		X		X

Notes: Each model has the younger sibling's consumption as the outcome and the older sibling's consumption as the covariate of interest. Model 1 uses the contemporaneous consumption of the older sibling and excludes sibling pairs in which the older sibling was interviewed after the younger sibling. Model 2 uses the lagged (prior survey year) consumption. All models are estimated via OLS with cluster robust standard errors at the household level and are limited to households with exactly two NLSY97 respondent siblings, in which the older sibling is 23 years old or younger. All models include a vector of controls for both siblings similar to those described in [Section 2](#). Means of each outcome (for younger siblings) are shown for each model. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 3: ATUS Sample Summary Statistics

	Mean	SD	Min	Max
<u>Respondent Characteristics:</u>				
Age	17.04	(1.72)	15.0	22.0
Female	0.49	(0.50)	0.0	1.0
In high school	0.60	(0.49)	0.0	1.0
In college	0.17	(0.38)	0.0	1.0
Employed	0.37	(0.48)	0.0	1.0
Surveyed on a weekend or holiday	0.30	(0.46)	0.0	1.0
Age of closest older sibling	19.70	(1.88)	16.0	23.0
Number of older siblings	1.15	(0.40)	1.0	4.0
<u>Minutes Spent (during ATUS-diary day):</u>				
At home	335.38	(220.72)	0.0	1440.0
At school	171.58	(226.34)	0.0	1020.0
At work	63.28	(158.08)	0.0	890.0
Alone	219.71	(196.14)	0.0	1370.0
With an older sibling	111.28	(176.54)	0.0	1200.0
With a parent	148.18	(192.14)	0.0	1200.0
With a friend	131.51	(205.64)	0.0	1140.0
With anyone else	184.67	(226.47)	0.0	1210.0
N	3,158			

Notes: Summary statistics from time-diaries for all ATUS respondents in the 2003-2019 waves who lived with one older sibling 23 years old or younger. All summary statistics are calculated using survey weights. Data retrieved from Hofferth et al. (2020).

Table 4: Discontinuities in Older Sibling Alcohol Consumption

Count of Drinking Days					
	(1)	(2)	(3)	(4)	(5)
Age 21+	1.447*** (0.152)	1.585*** (0.169)	1.834*** (0.241)	1.339*** (0.133)	1.328*** (0.146)
Constant	4.801*** (0.727)	3.205*** (0.502)	4.658*** (0.733)	3.720*** (0.083)	4.785*** (0.723)
Observations	16682	16682	16682	21376	17049
Any Drinking Days					
Age 21+	0.083*** (0.013)	0.090*** (0.013)	0.105*** (0.019)	0.083*** (0.011)	0.079*** (0.012)
Constant	0.545*** (0.057)	0.434*** (0.041)	0.536*** (0.058)	0.585*** (0.008)	0.556*** (0.056)
Observations	16682	16682	16682	21376	17049
Count of Binge Drinking Days					
Age 21+	0.454*** (0.100)	0.587*** (0.111)	0.678*** (0.161)	0.404*** (0.089)	0.405*** (0.095)
Constant	2.406*** (0.463)	1.496*** (0.322)	2.282*** (0.464)	1.667*** (0.051)	2.437*** (0.469)
Observations	16601	16601	16601	21268	16964
Any Binge Drinking Days					
Age 21+	0.057*** (0.012)	0.068*** (0.013)	0.073*** (0.020)	0.058*** (0.011)	0.050*** (0.012)
Constant	0.427*** (0.053)	0.305*** (0.041)	0.423*** (0.054)	0.342*** (0.007)	0.422*** (0.053)
Observations	16601	16601	16601	21268	16964
Fixed Effects	X		X	X	X
Quadratic			X		
Controls	X	X	X		X
Donut	X	X	X	X	

Notes: Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year. All models are estimated in a sample of NLSY97 respondents who are the oldest siblings currently residing in their household who are between the ages of 19 and 23. All models include cluster robust standard errors at individual level. Age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 5: Smoothness of Covariates at Cutoff

	Predicted Outcomes			
	Drinking Days	Any Drinking Days	Binge Drinking Days	Any Binge Binge Days)
Constant	2.415*** (0.091)	0.450*** (0.010)	1.143*** (0.060)	0.255*** (0.009)
Sibling 21+	-0.023 (0.084)	-0.004 (0.009)	0.008 (0.053)	-0.001 (0.008)
<i>N</i>	4373	4373	4369	4369

Notes: Outcomes are predicted values from a regression of the relevant consumption measure on the month and year of the survey, the race of the respondent, the gender of the respondent and siblings, educational attainment and enrollment of the respondent, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year. Results shown are from regressions of these predicted values on the age of the older sibling in months, an indicator for whether the older sibling is over 21, and their interaction. Standard errors are bootstrapped. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 6: Reduced Form

Count of Drinking Days					
	(1)	(2)	(3)	(4)	(5)
Sibling 21+	-0.137 (0.249)	0.098 (0.249)	-0.288 (0.431)	0.156 (0.229)	-0.058 (0.241)
Constant	2.577 (2.023)	4.555*** (0.830)	2.505 (2.008)	2.389*** (0.150)	2.885 (1.890)
Observations	4282	4282	4282	5527	4373
Any Drinking Days					
Sibling 21+	-0.031 (0.028)	-0.009 (0.028)	-0.059 (0.046)	0.004 (0.025)	-0.035 (0.027)
Constant	0.828*** (0.171)	0.657*** (0.089)	0.840*** (0.173)	0.459*** (0.016)	0.904*** (0.163)
Observations	4282	4282	4282	5527	4373
Count of Binge Drinking Days					
Sibling 21+	-0.337* (0.159)	-0.176 (0.169)	-0.372 (0.267)	-0.195 (0.161)	-0.297+ (0.153)
Constant	1.512 (1.485)	2.846*** (0.599)	1.440 (1.465)	1.313*** (0.107)	1.567 (1.383)
Observations	4278	4278	4278	5521	4369
Any Binge Drinking Days					
Sibling 21+	-0.025 (0.025)	0.000 (0.025)	-0.038 (0.040)	-0.017 (0.022)	-0.025 (0.024)
Constant	0.408* (0.183)	0.439*** (0.078)	0.402* (0.185)	0.282*** (0.015)	0.443* (0.184)
Observations	4278	4278	4278	5521	4369
FE	X		X	X	X
Quadratic			X		
Controls	X	X	X		X
Donut	X	X	X	X	

Notes: Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year. All models are estimated in a sample of NLSY97 respondents who have exactly one older sibling in their household, and where that older sibling is between the ages of 19 and 23. All models include cluster robust standard errors at individual level. Sibling age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 7: Reduced Form in Subsamples, Binge Days

	Parental Education		Sibling Sex Composition	
	Low (either < HS)	High (both ≥ HS)	Same Sex Siblings	Different Sex Siblings
Sibling 21+	-0.124 (0.277)	-0.417 ⁺ (0.217)	-0.616 ^{**} (0.231)	-0.040 (0.223)
Constant	-1.695 (1.476)	2.791 (2.454)	-0.194 (1.812)	0.614 (1.716)
Observations	1045	2206	2379	1899
	Older Sibling's Sex		Sibling Age Diff	
	Male Older Sibling	Female Older Sibling	≥ 30 Months	< 30 Months
Sibling 21+	-0.662 ^{**} (0.245)	0.064 (0.208)	-0.439 [*] (0.202)	-0.290 (0.261)
Constant	-6.995 [*] (2.725)	-0.268 (1.087)	1.284 (1.583)	13.177 ^{***} (3.087)
Observations	2266	2012	2312	1966
	Experience with Alcohol (younger sibling in prior year)		Experience with Alcohol (younger sibling in prior year)	
	Ever Drinkers	Never Drinkers	Ever Binge Drinkers	Never Binge Drinkers
Sibling 21+	-0.488 [*] (0.243)	-0.166 (0.117)	-0.644 (0.560)	-0.195 ⁺ (0.112)
Constant	-0.325 (1.845)	0.750 (0.483)	0.869 (2.909)	1.005 (0.672)
Observations	2511	1492	1069	2783

Notes: Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year. All models are estimated in the corresponding subgroup of a sample of NLSY97 respondents who have exactly one older sibling in their household, and where that older sibling is between the ages of 19 and 23. All models include cluster robust standard errors at individual level. Sibling age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 8: Reduced Form in Subsamples, Any Binge Days

	Parental Education		Sibling Sex Composition	
	Low (either < HS)	High (both ≥ HS)	Same Sex Siblings	Different Sex Siblings
Sibling 21+	-0.046 (0.057)	-0.009 (0.034)	-0.046 (0.038)	-0.012 (0.038)
Constant	-0.000 (0.305)	0.645** (0.239)	0.011 (0.196)	0.795 (0.540)
Observations	1045	2206	2379	1899
	Older Sibling's Sex		Sibling Age Diff	
	Male Older Sibling	Female Older Sibling	≥ 30 Months	< 30 Months
Sibling 21+	-0.039 (0.035)	-0.009 (0.038)	-0.020 (0.035)	-0.018 (0.037)
Constant	-2.136*** (0.254)	0.436 (0.370)	0.979* (0.408)	0.615* (0.312)
Observations	2266	2012	2312	1966
	Experience with Alcohol (younger sibling in prior year)		Experience with Alcohol (younger sibling in prior year)	
	Ever Drinkers	Never Drinkers	Ever Binge Drinkers	Never Binge Drinkers
Sibling 21+	-0.014 (0.039)	-0.043 ⁺ (0.023)	-0.002 (0.060)	-0.036 (0.027)
Constant	0.324 (0.204)	0.572 ⁺ (0.308)	-0.000 (0.451)	0.475** (0.183)
Observations	2511	1492	1069	2783

Notes: Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year. All models are estimated in the corresponding subgroup of a sample of NLSY97 respondents who have exactly one older sibling in their household, and where that older sibling is between the ages of 19 and 23. All models include cluster robust standard errors at individual level. Sibling age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 9: Alternative Samples

	Drinking Days	Any Drinking Days	Binge Drinking Days	Any Binge Drinking Days
Two oldest siblings living in the same household				
Sibling 21+	-0.137 (0.249)	-0.031 (0.028)	-0.337* (0.159)	-0.025 (0.025)
Constant	2.577 (2.023)	0.828*** (0.171)	1.512 (1.485)	0.408* (0.183)
Observations	4282	4282	4278	4278
Two oldest siblings				
Sibling 21+	0.026 (0.229)	-0.025 (0.026)	-0.252+ (0.147)	-0.024 (0.023)
Constant	5.618** (1.861)	0.659*** (0.195)	2.205* (1.069)	0.580* (0.247)
Observations	5100	5100	5094	5094
Two siblings living in the same household				
Sibling 21+	-0.230 (0.186)	-0.015 (0.020)	-0.202+ (0.121)	-0.026 (0.018)
Constant	2.232 (1.709)	0.451*** (0.104)	0.564 (1.139)	0.296** (0.099)
Observations	6988	6988	6975	6975
Two siblings				
Sibling 21+	-0.079 (0.161)	-0.025 (0.016)	-0.107 (0.101)	-0.020 (0.015)
Constant	2.354+ (1.218)	0.190* (0.077)	0.248 (0.820)	0.525*** (0.074)
Observations	9975	9975	9958	9958

Notes: The first panel is estimated in a sample of NLSY97 respondents who have exactly one older sibling in their household, and where that older sibling is between the ages of 19 and 23. The second panel is estimated in a sample of NLSY97 respondents who are the second oldest siblings, where their older sibling is between the ages of 19 and 23, and the older sibling may not currently reside in the same household as the respondent. The third panel is estimated in a sample of NLSY97 respondents who have one or more older siblings in their household where the closest older sibling to the respondent is between the ages of 19 and 23. The last panel is estimated in a sample of NLSY respondents who have one or more older siblings where the closest older sibling to the respondent is between the ages of 19 and 23, and the closest older sibling may not currently reside in the same household as the respondent. Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether the respondent has children, and an indicator for whether the respondent worked in the past year. All models include cluster robust standard errors at individual level. Sibling age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 10: Switcher and Non-Switcher Characteristics

	Switchers mean (sd)	Non-switchers mean (sd)	p-value
Age difference between the siblings	2.81 (1.36)	3.01 (1.65)	0.00
Siblings have same gender	0.55 (0.50)	0.53 (0.50)	0.04
<u>Younger Sibling's Characteristics:</u>			
Female	0.47 (0.50)	0.48 (0.50)	0.44
Race: Black	0.27 (0.44)	0.24 (0.43)	0.04
Race: Hispanic	0.24 (0.43)	0.24 (0.42)	0.75
Urban	0.78 (0.41)	0.75 (0.43)	0.01
AFQT score	46,037.04 (29,391.96)	42,094.33 (28,584.32)	0.00
Household Income	\$66,654.55 (57,999.39)	\$52,071.82 (48,741.99)	0.00
AFQT score	\$46,037.04 (29,391.96)	\$42,094.33 (28,584.32)	0.00
N	3716	1859	
N Sibling Pairs	1180	1423	

Notes: FE estimates of the effect of an older sibling turning 21 are identified only by groups who have variation in this treatment, i.e., younger siblings who show up in our sample at least once before their older sibling turns 21 and at least once after. This table compares these “switcher” sibling pairs to the corresponding “non-switcher” sibling pairs, in our main analysis sample (younger sibling is a NLSY-respondent, older sibling is between the ages of 19 and 23, siblings are the two oldest siblings in the household). The third column displays the p-value for the null hypothesis that the means in the switcher and non-switcher groups are equivalent.

Table 11: Switcher and Non-Switcher Characteristics in an Alternative Sample

	Switchers mean (sd)	Non-switchers mean (sd)	p-value
Age difference between the siblings	2.65 (1.32)	4.03 (2.10)	0.00
Siblings have same gender	0.52 (0.50)	0.53 (0.50)	0.68
<u>Younger Sibling's Characteristics:</u>			
Female	0.50 (0.50)	0.47 (0.50)	0.18
Race: Black	0.25 (0.43)	0.25 (0.43)	0.93
Race: Hispanic	0.20 (0.40)	0.28 (0.45)	0.00
Urban	0.76 (0.43)	0.76 (0.43)	0.76
AFQT score	45,620.70 (28,991.13)	40,877.99 (26,712.87)	0.00
Household Income	\$53,699.29 (56,351.16)	\$47,828.96 (48,448.64)	0.09
N	4240	345	
N Sibling Pairs	1788	278	

Notes: FE estimates of the effect of an older sibling turning 21 are identified only by groups who have variation in this treatment, i.e., younger siblings who show up in our sample at least once before their older sibling turns 21 and at least once after. This table compares these “switcher” sibling pairs to the corresponding “non-switcher” sibling pairs, in an alternative sample that does not require the siblings to live together (i.e., here the younger sibling is a NLSY-respondent, the older sibling is between the ages of 19 and 23, and the siblings are the two oldest siblings in the family). The third column displays the p-value for the null hypothesis that the means in the switcher and non-switcher groups are equivalent.

Table 12: Effect of Younger Sibling on Older Sibling

	Drinking Days	Any Drinking Days	Binge Drinking Days	Any Binge Drinking Days
Two youngest siblings living in the same household				
Sibling 21+	-0.117 (0.467)	0.003 (0.034)	0.060 (0.258)	0.055 (0.036)
Constant	12.259*** (2.778)	0.236 (0.260)	0.620 (1.550)	0.122 (0.260)
Observations	2380	2380	2333	2333
Two youngest siblings				
Sibling 21+	-0.013 (0.325)	0.016 (0.026)	-0.016 (0.214)	0.062* (0.029)
Constant	10.813** (3.939)	0.513** (0.177)	2.064+ (1.158)	0.152 (0.277)
Observations	3758	3758	3702	3702
Two siblings living in the same household				
Sibling 21+	0.006 (0.227)	0.007 (0.017)	-0.112 (0.136)	0.001 (0.018)
Constant	4.968*** (1.420)	0.648*** (0.119)	2.764** (1.032)	0.353* (0.139)
Observations	7557	7557	7398	7398
Two siblings				
Sibling 21+	-0.056 (0.163)	0.002 (0.013)	0.000 (0.101)	0.011 (0.013)
Constant	5.380*** (1.022)	0.670*** (0.098)	2.464*** (0.728)	0.362*** (0.097)
Observations	13367	13367	13145	13145

Notes: The first panel is estimated in a sample of NLSY97 respondents who have exactly one younger sibling in their household, and where that younger sibling is between the ages of 19 and 23. The second panel is estimated in a sample of NLSY97 respondents who are the second youngest siblings, where their younger sibling is between the ages of 19 and 23, and the younger sibling may not currently reside in the same household as the respondent. The third panel is estimated in a sample of NLSY97 respondents who have one or more younger siblings in their household where the closest younger sibling to the respondent is between the ages of 19 and 23. The last panel is estimated in a sample of NLSY respondents who have one or more younger siblings, where the closest younger sibling to the respondent is between the ages of 19 and 23, and the closest younger sibling may not currently reside in the same household as the respondent. Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether the respondent has children, and an indicator for whether the respondent worked in the past year. All models include cluster robust standard errors at individual level. Sibling age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 13: Parenting style changes at the cutoff

	Degree of Parental Monitoring				
	(1)	(2)	(3)	(4)	(5)
Sibling 21+	-0.243 (0.321)	-0.722* (0.354)	-0.311 (0.508)	-0.421 (0.292)	-0.198 (0.294)
Constant	11.307*** (2.290)	9.525*** (1.418)	11.337*** (2.324)	8.653*** (0.167)	9.482*** (2.353)
Observations	1467	1467	1467	1937	1493
	Parents: Authoritarian/Authoritative				
	(1)	(2)	(3)	(4)	(5)
Sibling 21+	-0.010 (0.025)	-0.000 (0.023)	-0.017 (0.041)	0.007 (0.024)	-0.010 (0.024)
Constant	0.348 (0.364)	-0.135* (0.069)	0.369 (0.369)	0.262*** (0.016)	0.347 (0.368)
Observations	4317	4317	4317	5575	4408

Notes: Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year. All models are estimated in a sample of NLSY97 respondents who are the second oldest siblings currently residing in their household, where the oldest sibling is between the ages of 19 and 23. All models include cluster robust standard errors at individual level. Sibling age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

Table 14: Continuity Models

Count of Binge Drinking Days				
Sibling 21+	-0.429*** [-0.598,-0.261]	-0.355** [-0.529,-0.180]	-0.118 [-0.612,0.372]	-0.129 [-0.612, 0.597]
N	2874	4124	3931	4124
Bandwidth	16.64	24.00	22.86	24.00
Any Binge Drinking Days				
Sibling 21+	-0.027* [-0.048,-0.005]	-0.029* [-0.056,-0.002]	-0.000 [-0.061,0.075]	-0.000 [-0.069,0.107]
N	2555	4124	4278	4124
Bandwidth	14.07	24.00	24.59	24.00
FE	X	X		
Quadratic				
Controls	X	X	X	X
Donut	X	X	X	X

Notes: Controls include the month and year of the survey, the gender and age of the respondent and siblings, race of respondents, educational attainment and enrollment of respondents, geography (urban/rural, census region), household size, AFQT score of the respondent, an indicator for whether or not the respondent has children, and an indicator for whether the respondent worked in the past year. All models include the MSE-optimal estimates and bias-corrected robust 95% confidence intervals where the standard errors are cluster robust at individual level. Models 1 and 3 use MSE-optimal bandwidths, and the other models use our ad-hoc 24 month bandwidth. Sibling age is centered at 21 years. +, *, **, and *** denote statistical significance at the 10%, 5%, 1%, and 0.1% levels, respectively.